

Exploratory Data Analysis of Audi Vehicle Listings

A Data Analytics Project Report

1. Introduction

Used-car pricing is influenced by a mix of vehicle age, mileage, specification, and brand positioning. This project analyzes a dataset of Audi vehicle listings to understand how these factors interact and to surface patterns that could support a future price-prediction model. The analysis was carried out in Python (pandas, matplotlib, seaborn) in a Google Colab notebook, covering data cleaning, exploratory data analysis (EDA), visualization, and correlation/outlier analysis.

2. Objective

- Understand the structure, quality, and distribution of the Audi listings dataset.
- Visualize how price relates to year, mileage, engine size, transmission, and fuel type.
- Identify the strongest numeric drivers of price and quantify outliers.
- Translate the findings into practical recommendations for a downstream price-prediction pipeline.

3. Dataset Description

The source file, cars_dataset.csv, contains **72,435 rows** across **10 columns**: model, year, price, transmission, mileage, fuelType, tax, mpg, engineSize, and Make. There were no missing values or column-level data-quality issues beyond stray whitespace in text fields, which was stripped during cleaning.

Data-quality note: although the file was described as an Audi-only export, the Make column shows it actually spans **seven brands** — Audi, BMW, Ford, VW, Toyota, Skoda, and Hyundai — likely bundled together from the same public UK used-car source. The analysis below filters to the **10,668 Audi listings** to match the project's stated scope; a brief cross-brand comparison is included in Section 7 since the data supported it.

Field	Description
model	Audi model name (A1–A8, Q2–Q8, S/RS variants, TT, R8)
year	Registration year

Field	Description
price	Listing price (£)
transmission	Manual / Automatic / Semi-Auto
mileage	Recorded mileage
fuelType	Petrol / Diesel / Hybrid (a few Electric/Other rows in the full file)
tax	Annual road tax (£)
mpg	Fuel economy (miles per gallon)
engineSize	Engine displacement (litres)

4. Methodology

1. **Load & clean:** read the CSV, strip whitespace from headers and string fields, confirm no nulls or duplicate rows.
2. **Scope:** filter the dataset to Make == 'audi' (10,668 rows).
3. **Univariate analysis:** distribution plots (histograms, boxplots) for price, mileage, year, engine size, mpg, and tax; bar charts for categorical fields (model, transmission, fuel type).
4. **Bivariate analysis:** price plotted against year, mileage, engine size, transmission, and fuel type to identify relationships.
5. **Correlation analysis:** Pearson correlation matrix across all numeric fields, ranked by relationship to price.
6. **Outlier detection:** IQR method applied to price to flag unusually high/low listings.
7. **Cross-brand comparison:** a bonus summary using the full unfiltered dataset.

5. Key Findings

5.1 Price distribution

Audi listing prices range from **£1,490 to £145,000**, with a mean of **£22,897** and median of **£20,200**. The distribution is right-skewed — most listings cluster between £15,000 and £28,000 (the interquartile range), while a long tail of high-end R8, RS, and S-line models pulls the mean above the median.

5.2 Mileage

Mileage ranges from near-new to **323,000** miles, with a median of **19,000** miles. As expected for a used-car market, the bulk of listings sit under 40,000 miles.

5.3 Price by registration year

Average price rises steadily with newer registration years — from roughly **£4,650** for 1997-registered cars up to **£35,967** for 2020-registered cars — reflecting both depreciation and the general upward drift in new-Audi pricing over the period.

5.4 Correlation with price

Feature	Correlation with price
year	+0.59
engineSize	+0.59
tax	+0.36
mileage	-0.54
mpg	-0.60

Fuel economy (mpg) has the strongest single relationship with price, but in the **opposite direction expected from a "feature"** — it's better read as a proxy for engine size and model tier: larger, more powerful (and pricier) engines are simply less fuel-efficient. Year and engine size are the next-strongest positive drivers of price; mileage is the clearest negative driver, in line with normal depreciation.

5.5 Model-level patterns

- **Highest-volume models:** A3 (1,929 listings), Q3 (1,417), A4 (1,381), A1 (1,347), A5 (882), Q5 (877) — these make up the bulk of the used Audi market.
- **Highest average price:** R8 (**£97,652**), Q8 (**£60,115**), RS6 (**£55,964**), RS5 (**£51,265**), RS4 (**£50,152**), SQ7 (**£49,269**) — performance and flagship SUV models command a large premium despite far lower listing volumes.

5.6 Transmission and fuel type

Manual transmission is the most common (4,369 listings), followed by Semi-Automatic (3,591) and Automatic (2,708). Diesel (5,577) narrowly outnumbers Petrol (5,063) in this segment, with

Hybrid a small minority (28). Price differences across transmission and fuel type exist but are noticeably smaller than the effect of engine size or mileage.

5.7 Outliers

Using the IQR method, listings priced below **£4,158** or above **£47,279** are flagged as outliers. **443 listings (4.15%)** fall outside this range — almost entirely high-end RS/S-line, Q7/Q8, and R8 listings rather than data-entry errors, meaning they represent genuine premium-segment pricing rather than noise.

5.8 Cross-brand context (bonus)

Within the full unfiltered file, Audi has the **highest average listing price of the seven brands (£22,897)**, narrowly ahead of BMW (£22,733), and well above VW (£16,839), Skoda (£14,275), Hyundai (£12,750), Toyota (£12,522), and Ford (£12,280) — consistent with Audi and BMW's premium market positioning.

6. Conclusion

The EDA confirms intuitive used-car pricing dynamics within the Audi segment: **vehicle age, engine size, and mileage are the dominant drivers of price**, while transmission and fuel type play a secondary role. Performance (RS/S) and flagship (Q7/Q8/A8/R8) models occupy a distinct high-price tier that shows up clearly as statistical outliers, while everyday models (A1/A3/A4/Q3/Q5) form the bulk of the market around the £15k–£28k range.

7. Recommendations for Further Work

- **Modeling:** one-hot encode model, transmission, and fuelType; log-transform price and mileage to reduce skew before regression modeling.
- **Outlier handling:** consider modeling premium/performance models (RS/S/Q7/Q8/A8/R8) separately from mainstream models, since they follow a different price curve.
- **Feature engineering:** derive a car_age feature from year, and consider interaction terms between engineSize and model tier.
- **Data quality:** if the source file continues to bundle multiple makes, add a validation step that asserts Make == 'audi' before any Audi-specific analysis runs.

Analysis performed in Python (pandas, matplotlib, seaborn) on cars_dataset.csv; full code and visualizations are available in the accompanying Colab notebook, Audi_Cars_EDA.ipynb.